

# Map Visualization in Power BI

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The capacity to translate complex information into useful insights is crucial in today's data-driven environment. This potential can be unlocked through data visualization. We can make sense of enormous amounts of data, spot trends, and reach conclusions that direct our decision-making processes by visualizing the data. Map visualization is a powerful type of data visualization.

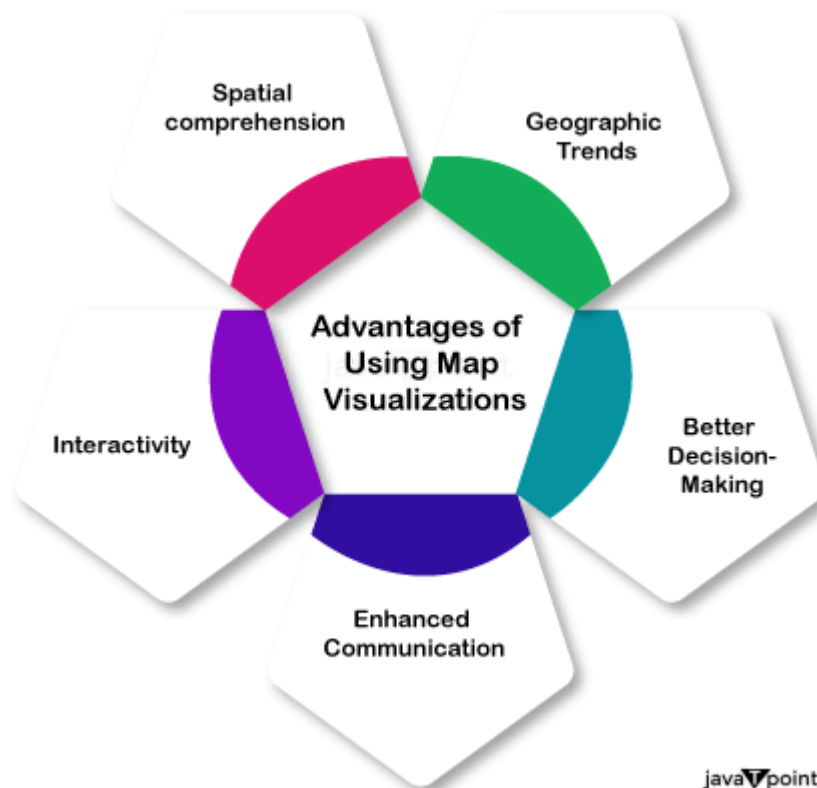
## **What are Map Visualisations in Power BI**

Power BI is a potent tool in the world of business intelligence and data analytics. In essence, Power BI enables users to leverage data and produce insightful findings. Map visualizations stand out to show data in a geographic context among its outstanding array of visualization possibilities. They enable us to investigate not only the "what" and "how" of our data but also its "where." Because so much of the world's data includes a geographic component, this geographical aspect is essential.

In Power BI, map visualizations are essentially ways to plot data points on a map. Depending on the kind of data they represent, they take on a variety of shapes. For instance, they can display the locations of your business's stores, the geographic areas where your products do the best business or even the routes taken by delivery trucks. Putting dots on a map is just the first step; the next steps involve identifying patterns, noticing trends, and making choices depending on the spatial context of your data. In industries like retail, logistics, healthcare, and many others, this is especially useful.

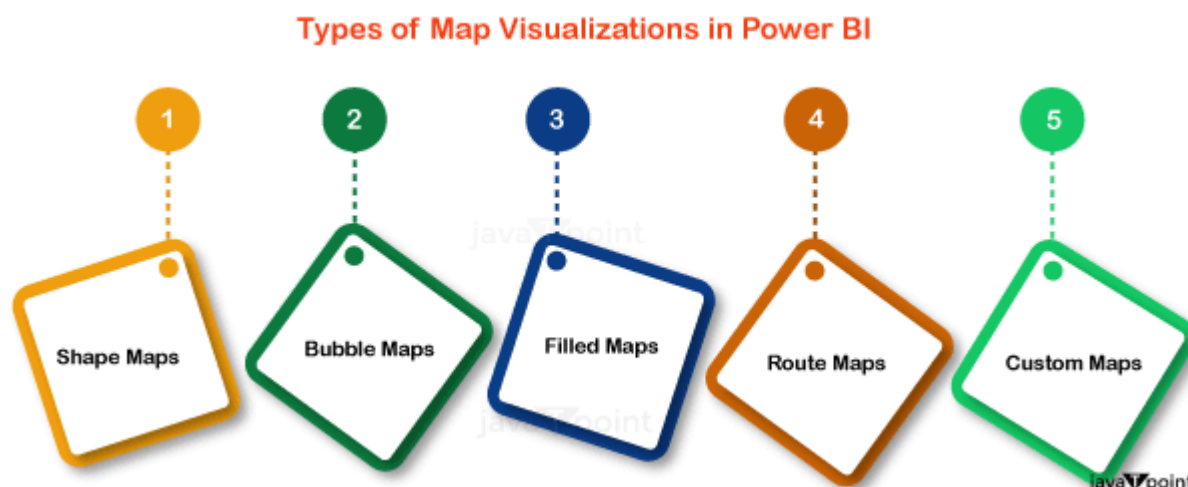
Visualizations of maps are essential for comprehending data in its geographic context. They can represent a range of geographic data types. They may represent particular locales, regional boundaries, or even dynamic aspects like routes. These visualizations are tools for uncovering hidden insights rather than just mapping out cities and nations. You will learn how to make the most of this useful tool in your data-driven decision-making processes as we go deeper into the realm of map visualizations in Power BI.

# Advantages of Using Map Visualizations



1. **Spatial comprehension:** The visual context that is frequently missing from conventional data tables and charts is provided by map visualizations. Whether it's the distribution of consumers, the location of assets, or the development of a disease, they help make complex spatial data intelligible. It is simpler to spot patterns and correlations in your data, thanks to the visual context.
2. **Geographic Trends:** Many industries require an understanding of regional trends. Retailers, for instance, can observe the hotspots for demand for their goods. Logistics businesses can optimize delivery routes. Medical professionals can locate disease hotspots. You may rapidly see these trends and take relevant action using map visualizations.
3. **Better Decision-Making:** Maps are logical decision-making aids. They assist in addressing location-based issues like where to set up a new store, which regions require more promotion, or how to distribute resources. Decision-makers can take geographic considerations into account when formulating strategies thanks to map visualizations.
4. **Enhanced Communication:** Everyone enjoys looking at maps. People speak of all backgrounds and cultures. Your results and recommendations will be easier to convey to a wide audience if you use map visualizations. They can understand what you're attempting to say really quickly.
5. **Interactivity:** In Power BI, a lot of map visualizations are interactive. Users can zoom in, choose map spots, and view more information. This level of interaction allows for deeper exploration and insights. It's particularly useful when you need to see fine-grained details within a broader geographical context.

# Types of Map Visualizations in Power BI



1. **Shape Maps:** Shape maps are ideal for displaying boundaries and regions. They can display nations, states, counties, and other governmental or political subdivisions. When your data is arranged according to geographical regions, these maps are employed. A shape map is the best option, for instance, if you want to show sales data broken down by state.
2. **Bubble Maps:** The use of bubble maps is excellent for displaying data that points of varied sizes can represent. These points could stand in for figures like population or sales volume. When you want to highlight the importance of particular locations, use bubble maps. Larger bubbles indicate greater values.
3. **Filled Maps:** Color is used in filled maps to depict data values within geographic regions. When your data is tied to geographic areas, and you want to show differences in a data point's magnitude throughout the map, you utilize them. For instance, a filled map can be used to display population density, with the colors denoting the distribution of individuals.
4. **Route Maps:** Maps of routes show the flow or links between various areas. They are helpful for showing tracks taken, such as aircraft trajectories or delivery routes. You may optimize processes by using route maps to understand better how entities move between places.
5. **Custom Maps:** Custom maps provide you the freedom to design map visualizations with any unique forms or non-traditional geographic locations. This enables you to modify the map to suit your requirements. For instance, you might show sales territories exclusive to your company on a bespoke map.

The choice will rely on your data and the narrative you want to convey. Each style of map visualization offers advantages over the others. Power BI offers the ability to generate the ideal map visualization for your data-driven projects, whether you're presenting regional sales trends, population densities, or supply chain operations.

## Using Power BI to Create a Map Visual

The tool for creating map visualizations in Power BI is a versatile and efficient way to display geographic data. Follow these steps to make a map graphic with position, legends, bubble size, and tooltips using a dataset of Products:

1. Start by examining the relevant columns in your dataset (for instance, product Product Category, Location, Salary Sum, Customer Name, Latitude, and Longitude).

## Create Table

|    | Location      | Product Category | Latitude | Longitude | Customer Na...  | Sum of Salary | + |
|----|---------------|------------------|----------|-----------|-----------------|---------------|---|
| 1  | New York      | Electronics      | 40.7128  | -74.0060  | John Smith      | 55000         |   |
| 2  | Los Angeles   | Clothing         | 34.0522  | -118.2437 | Lisa Johnson    | 48000         |   |
| 3  | Chicago       | Home Appliances  | 41.8781  | -87.6298  | David Brown     | 60000         |   |
| 4  | Houston       | Electronics      | 29.7604  | -95.3698  | Sarah Wilson    | 52000         |   |
| 5  | Miami         | Clothing         | 25.7617  | -80.1918  | Michael Lee     | 45000         |   |
| 6  | San Francisco | Home Appliances  | 37.7749  | -122.4194 | Emily Davis     | 58000         |   |
| 7  | Seattle       | Electronics      | 47.6062  | -122.3321 | Robert Garcia   | 52000         |   |
| 8  | Philadelphia  | Clothing         | 39.9526  | -75.1652  | Linda Martinez  | 49000         |   |
| 9  | Dallas        | Home Appliances  | 32.7767  | -96.7970  | James Clark     | 60000         |   |
| 10 | Phoenix       | Electronics      | 33.4484  | -112.0740 | Susan Turner    | 54000         |   |
| 11 | Boston        | Clothing         | 42.3601  | -71.0589  | William White   | 47000         |   |
| 12 | Denver        | Home Appliances  | 39.7392  | -104.9903 | Patricia Green  | 59000         |   |
| 13 | Atlanta       | Electronics      | 33.7490  | -84.3880  | Richard Harris  | 55000         |   |
| 14 | San Diego     | Clothing         | 32.7157  | -117.1611 | Elizabeth Smith | 48000         |   |
| 15 | Las Vegas     | Home Appliances  | 36.1699  | -115.1398 | Joseph Martinez | 61000         |   |
| 16 | Portland      | Electronics      | 45.5051  | -122.6750 | Maria Johnson   | 53000         |   |
| 17 | Minneapolis   | Clothing         | 44.9778  | -93.2650  | Daniel Davis    | 48000         |   |
| 18 | Detroit       | Home Appliances  | 42.3314  | -83.0458  | Jennifer Hall   | 59000         |   |
| 19 | Orlando       | Electronics      | 28.5383  | -81.3792  | Kevin Turner    | 54000         |   |
| 20 | Baltimore     | Clothing         | 39.2904  | -76.6122  | Karen Robinson  | 49000         |   |
| +  |               |                  |          |           |                 |               |   |

Name:

Load

Edit

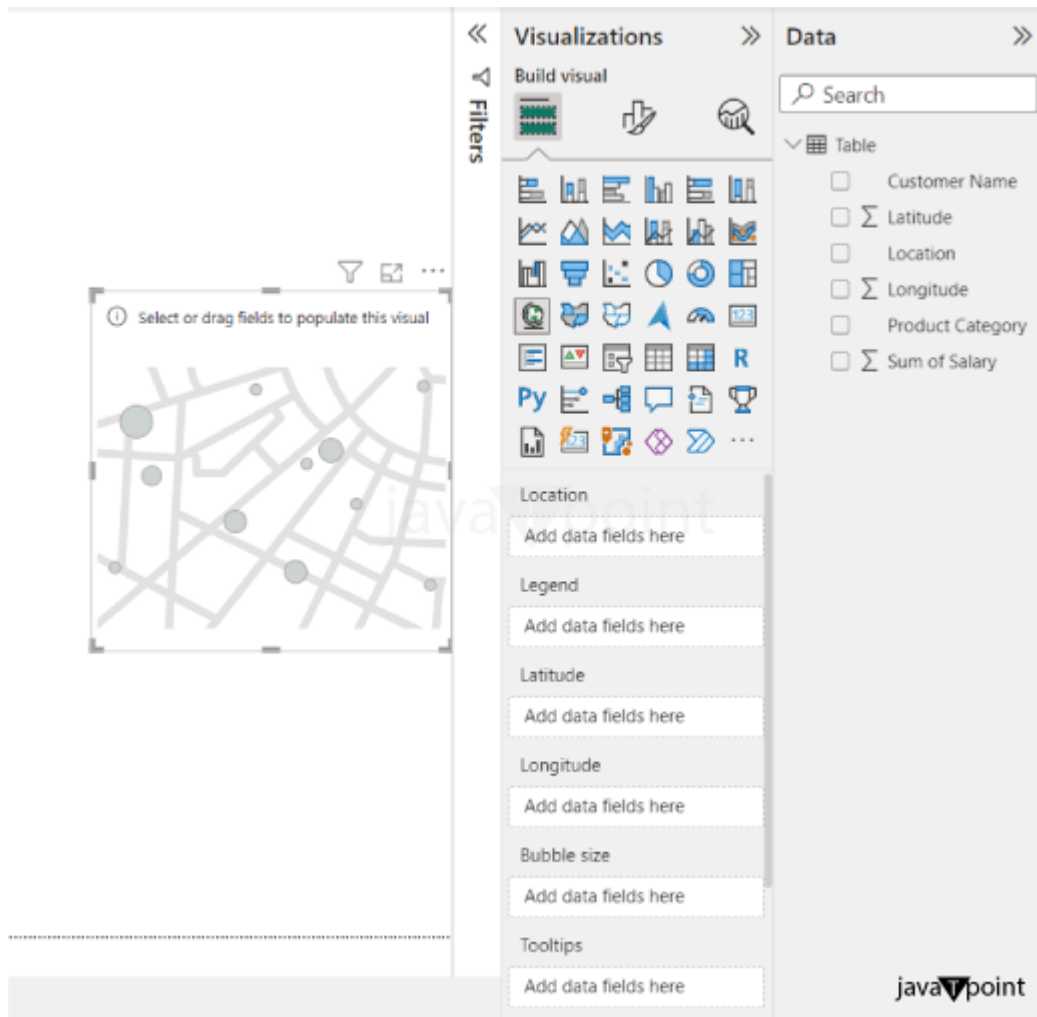
Cancel

java point

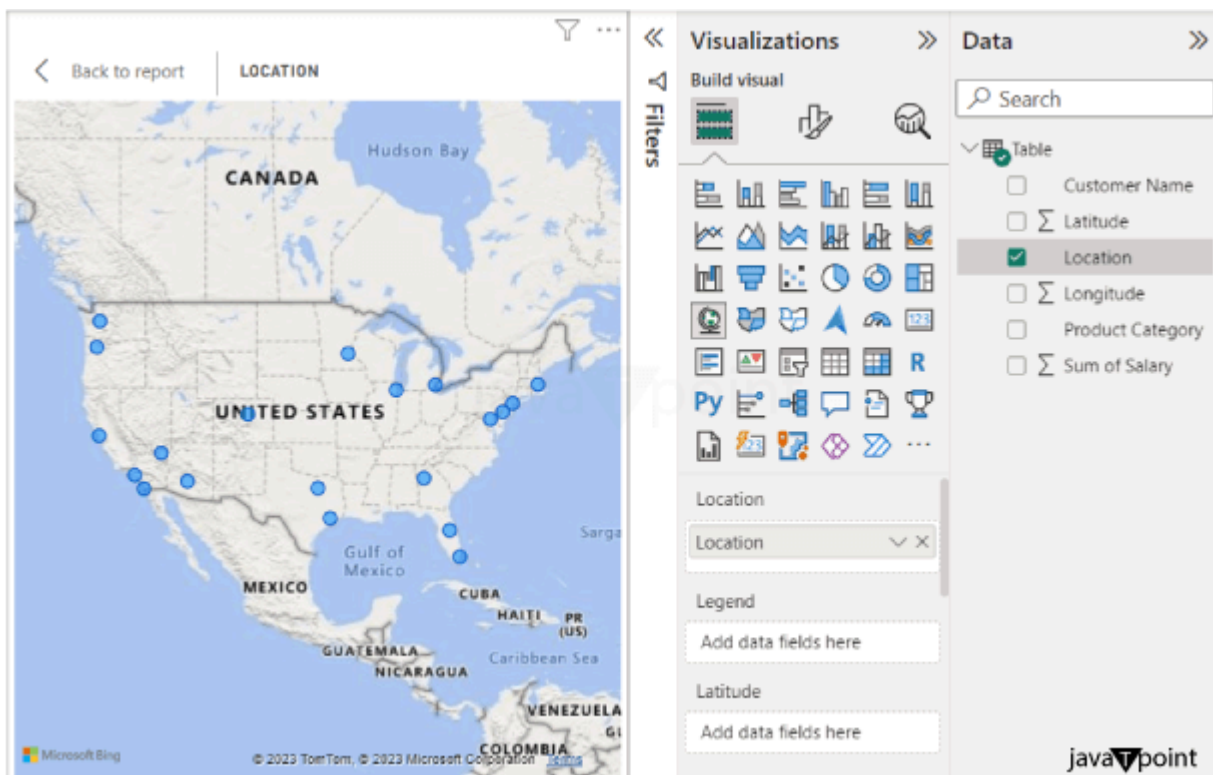
2. In the Power BI interface, go to the Visualisations section.
3. Select the "Map" icon to create a map-based graphic.

The screenshot shows the Power BI interface. On the left, the 'Visualizations' pane is open, displaying a grid of visualization icons. The 'Map' icon, which shows a map of the United States, is highlighted with a white tooltip that says 'Map'. Below the icons, there are sections for 'Values' (with a text box 'Add data fields here'), 'Drill through' (with a text box 'Add drill-through fields here'), and 'Cross-report' (with a toggle switch set to 'Off'). At the bottom, there is a 'Keep all filters' toggle switch set to 'On'. On the right, the 'Data' pane is visible, showing a table with columns: Customer Name, Latitude, Location, Longitude, Product Category, and Sum of Salary. A search bar is at the top of the Data pane.

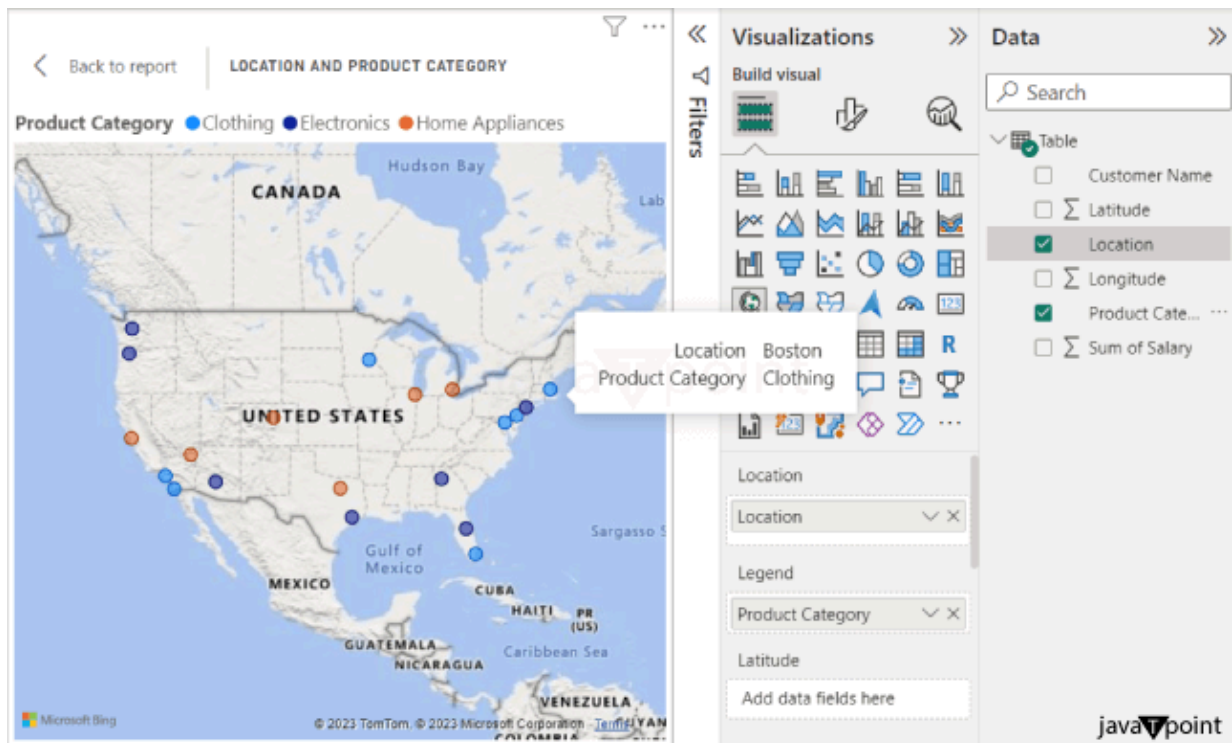
4. A blank map canvas will now show up. Now, let's add columns to the map.



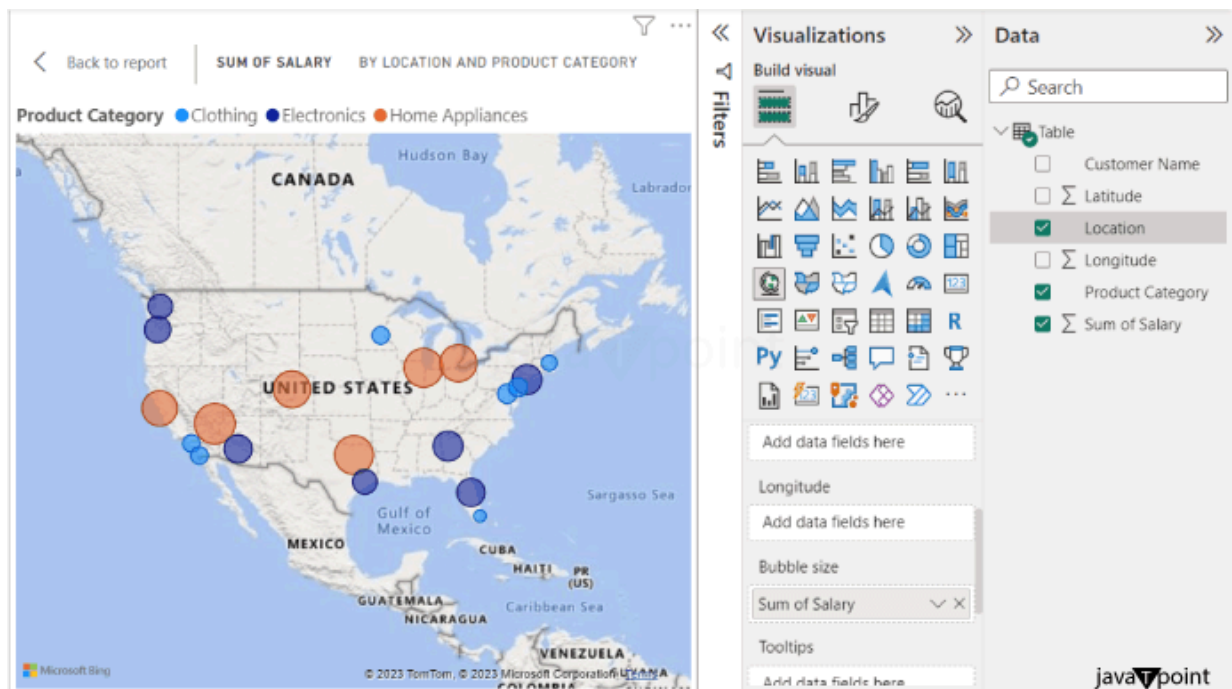
5. Drag and drop location information into the location field to add it to the map. The names of the cities in your dataset will be mapped using this. Remember that each city is originally represented by the same color (for instance, blue).



6. Sort your data into categories using Legends. Drag the "Product category" over to the "Legend" section. This action has given each group its hue. As an illustration, various colors will be used for apparel, gadgets, and home appliances, respectively.

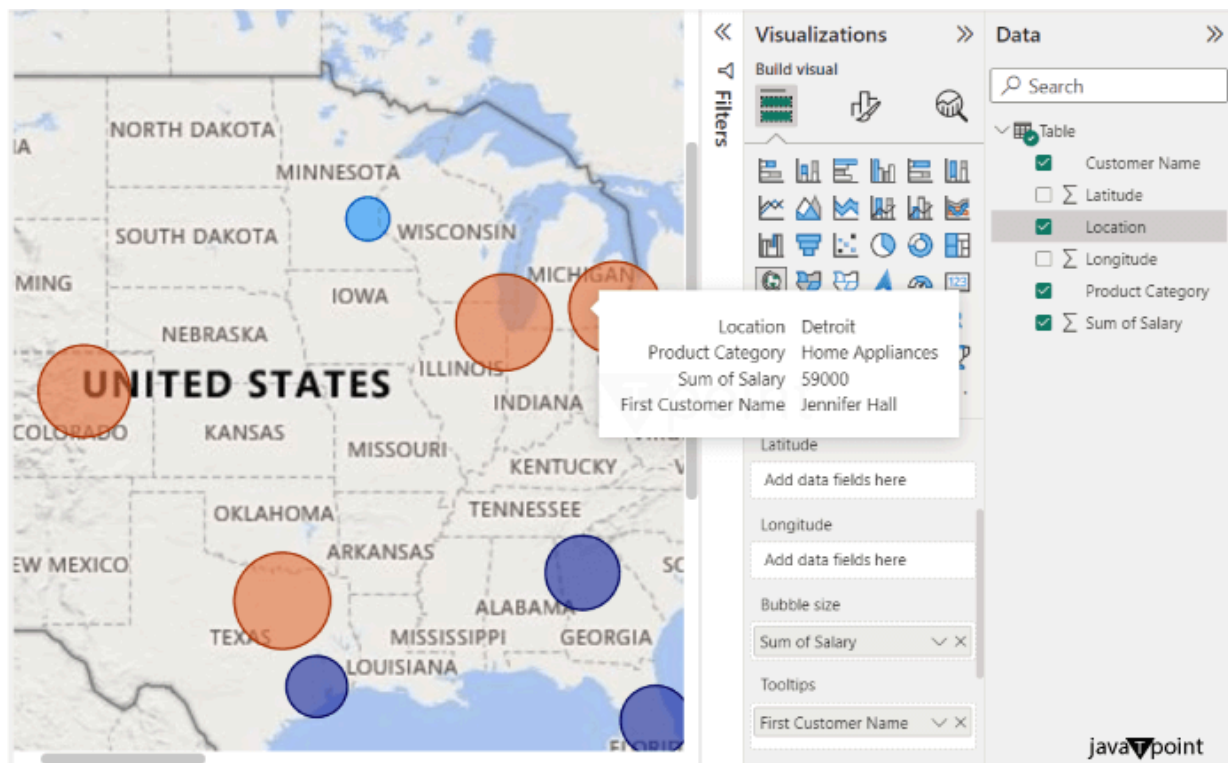


7. Different amounts may be indicated by different-sized bubbles on your map. To do this, drag and drop the Sum of Salary into the Bubble size area. The size of each bubble will depend on the related salary value.

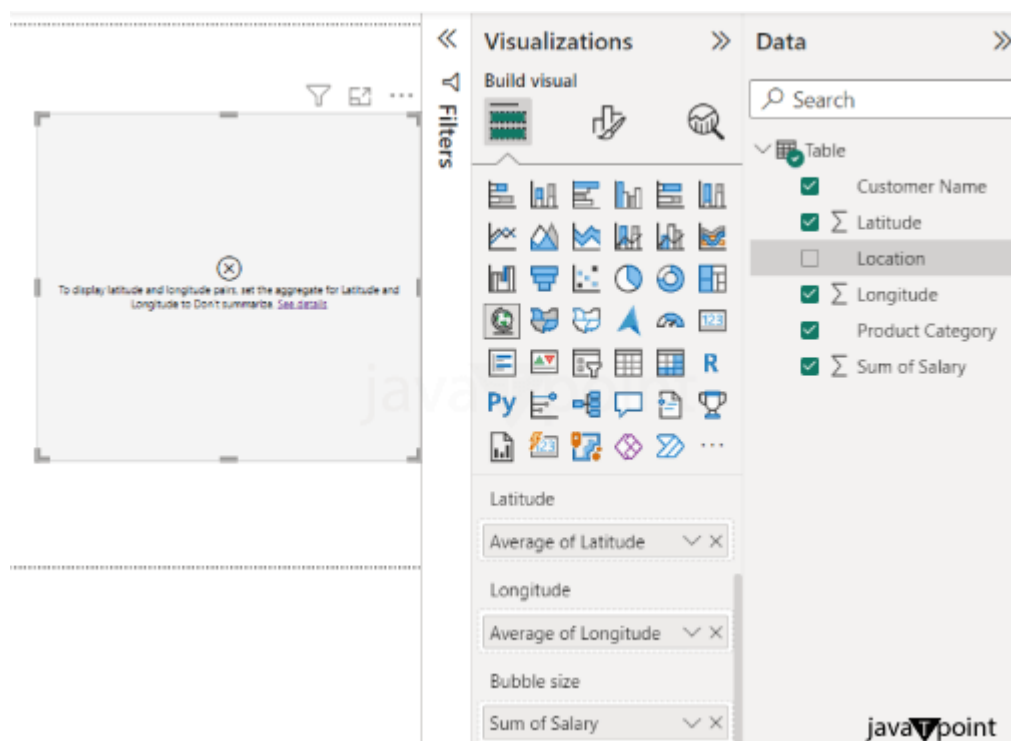


8. Now, let's add tooltips to the map. Tooltips provide more information when you hover your mouse over a data point. To include the Customer Name in your tooltips, drag & drop it there.



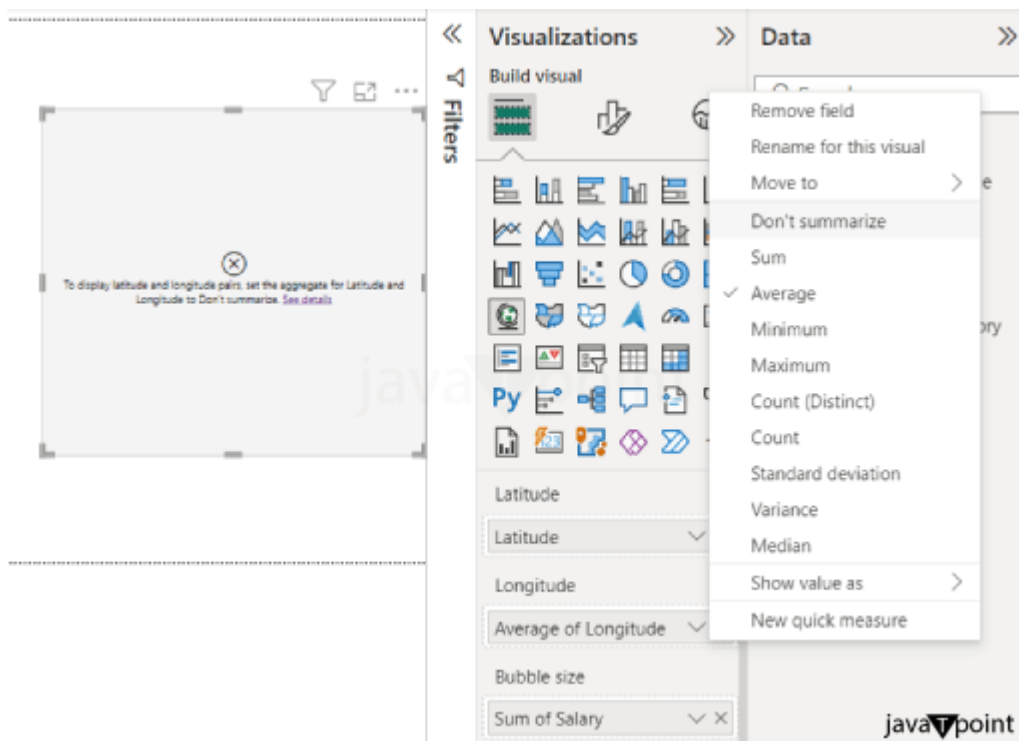


9. An alternative to the location is to include the latitude and longitude. Latitude and longitude are used to locate a certain location on the globe. To accomplish this, substitute Latitude and Longitude for Location in the Location section of the map. Power BI, by default, tries to summarise numerical values. Therefore, mistakes can happen.

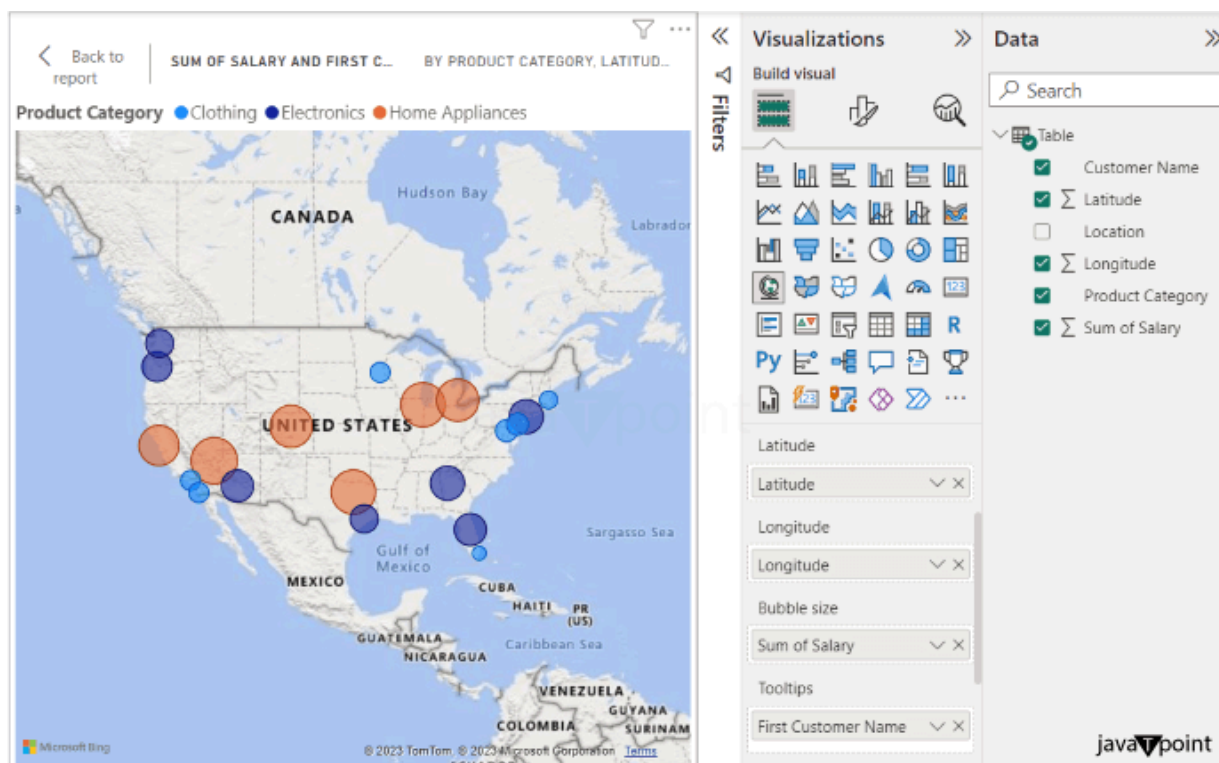


10. To resolve the problem, right-click on the Latitude and Longitude and select the "Don't summarise" option from the context menu. Power BI is unable to summarise these numbers.





11. When you have successfully finished the previous steps, you will have a map visual in Power BI that shows your data based on location, legends, bubble size, and tooltips.



## How Can a Map Be Formatted in Power BI?

It's important to look into the many formatting options after creating a map successfully in Power BI. This includes adding titles to the map, customizing its look and controls, changing the legend's features, editing bubble settings, and revising category names. Power BI offers two different types of formatting options: visual formatting and generic formatting.

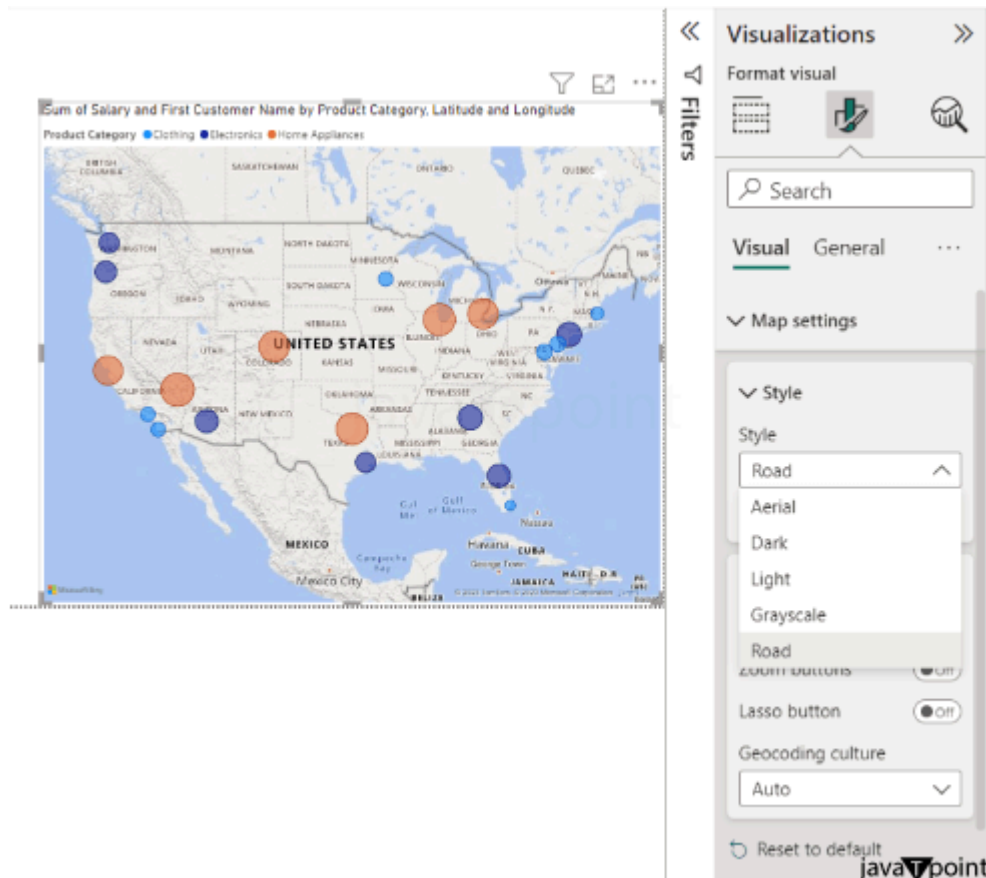
## Visual Formatting

Map Settings, Legends, Bubbles, and Category Labels are the four primary options for visual formatting. Let's look at each of these options to get a better idea of how you might format your map image.

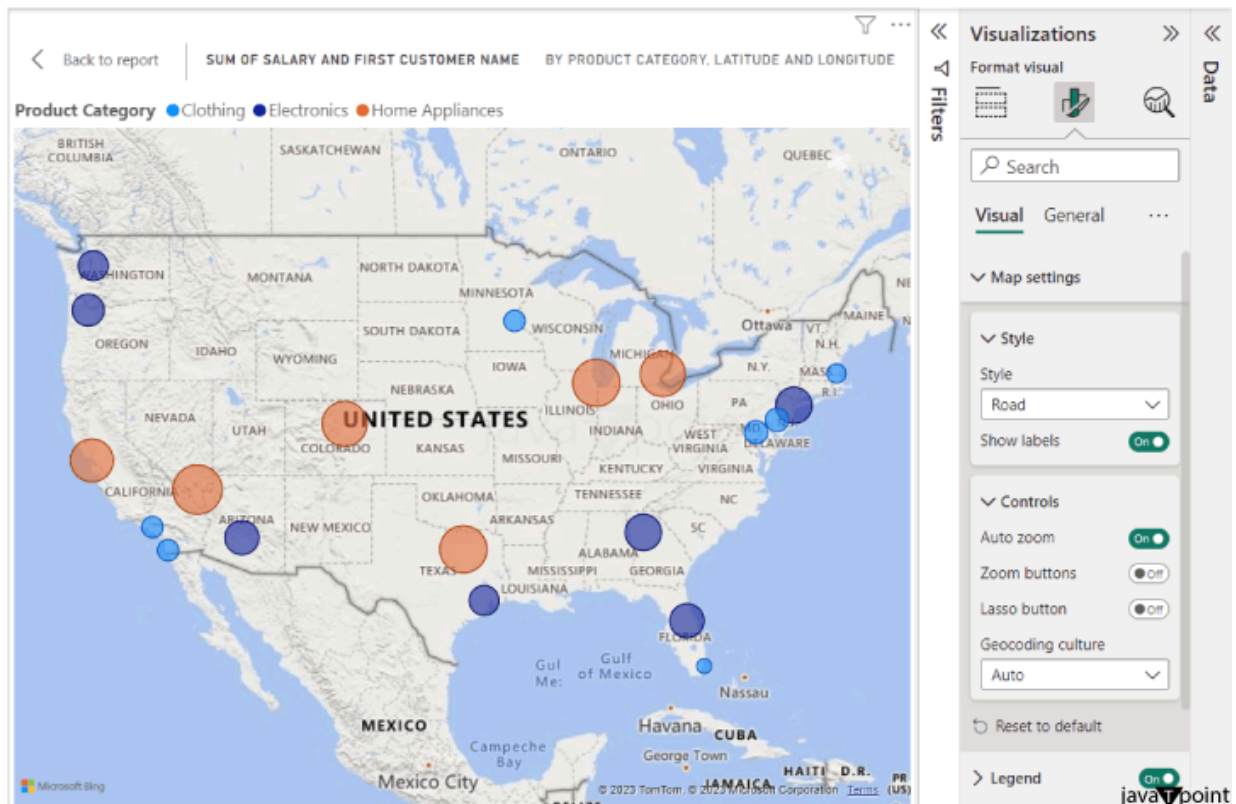
### Map Settings

The map settings allow you to change the zoom level, style, and overall appearance of the map.

1. Choose "Map Settings" from the drop-down menu. The two primary sub-options in the drop-down menu that appear are "Style" and "Controls."



2. Under the "Style" sub-option, you can choose from a range of map styles, including Aerial, Dark, LightScale, and Road.

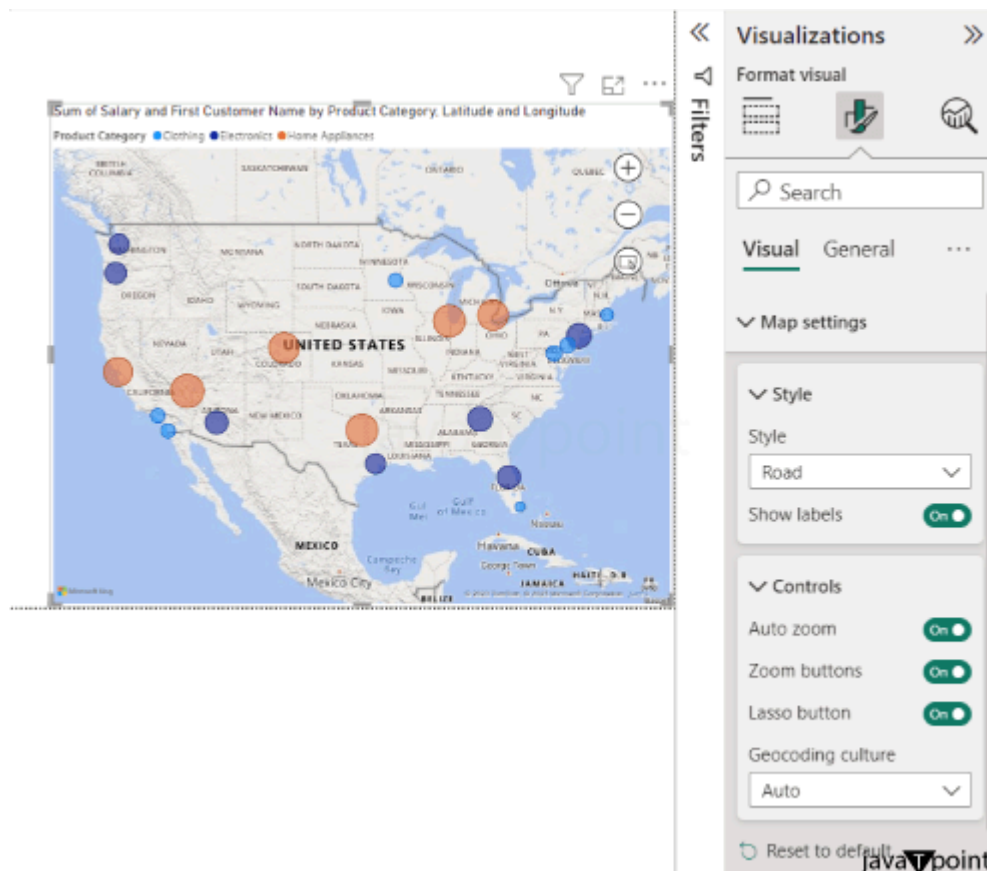


3. The "Controls" sub-option provides three sliders to modify various map features.

**Auto Zoom:** This slider automatically adjusts the zoom level to concentrate on the data points. It's the standard slider.

**Zoom Buttons:** By adding zoom in (+) and zoom out (-) buttons to the map, selecting this option enables manual zooming.

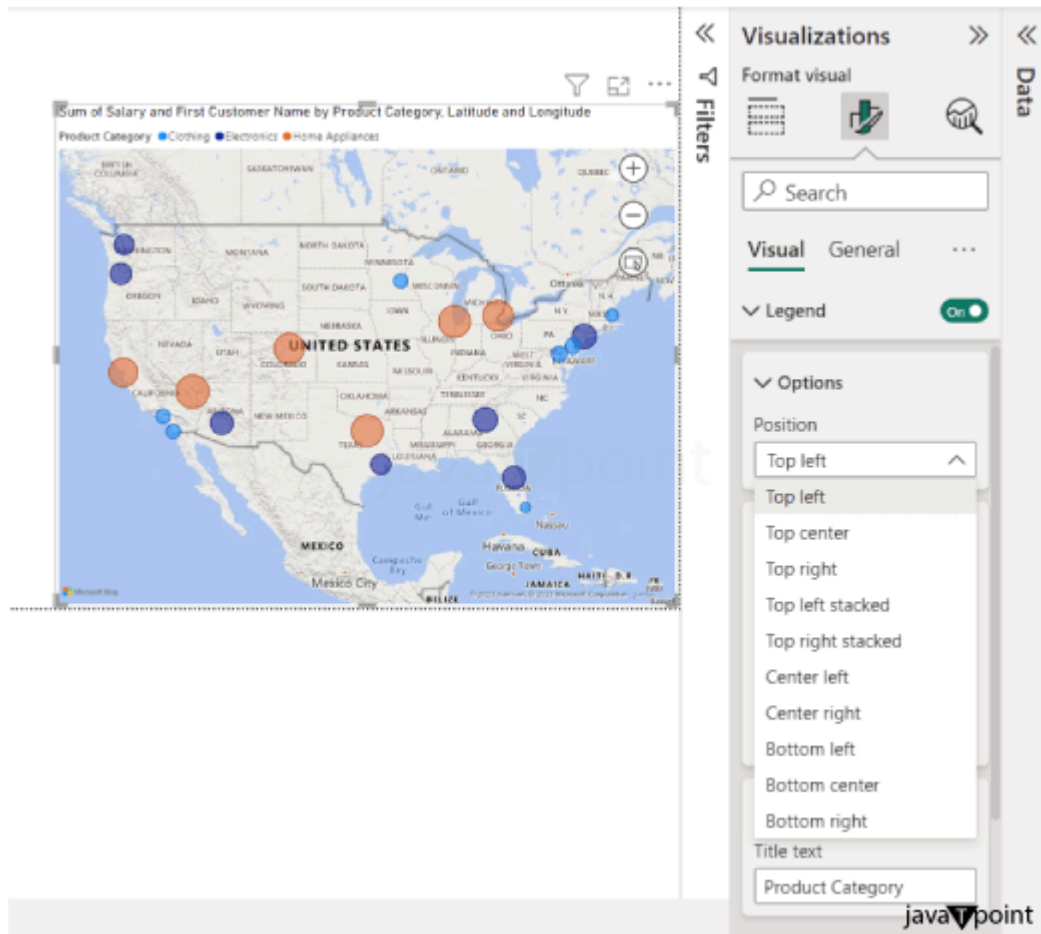
**Lasso Buttons:** When this feature is activated, a rectangular icon on the map appears, allowing you to select and focus on certain regions of the dataset.



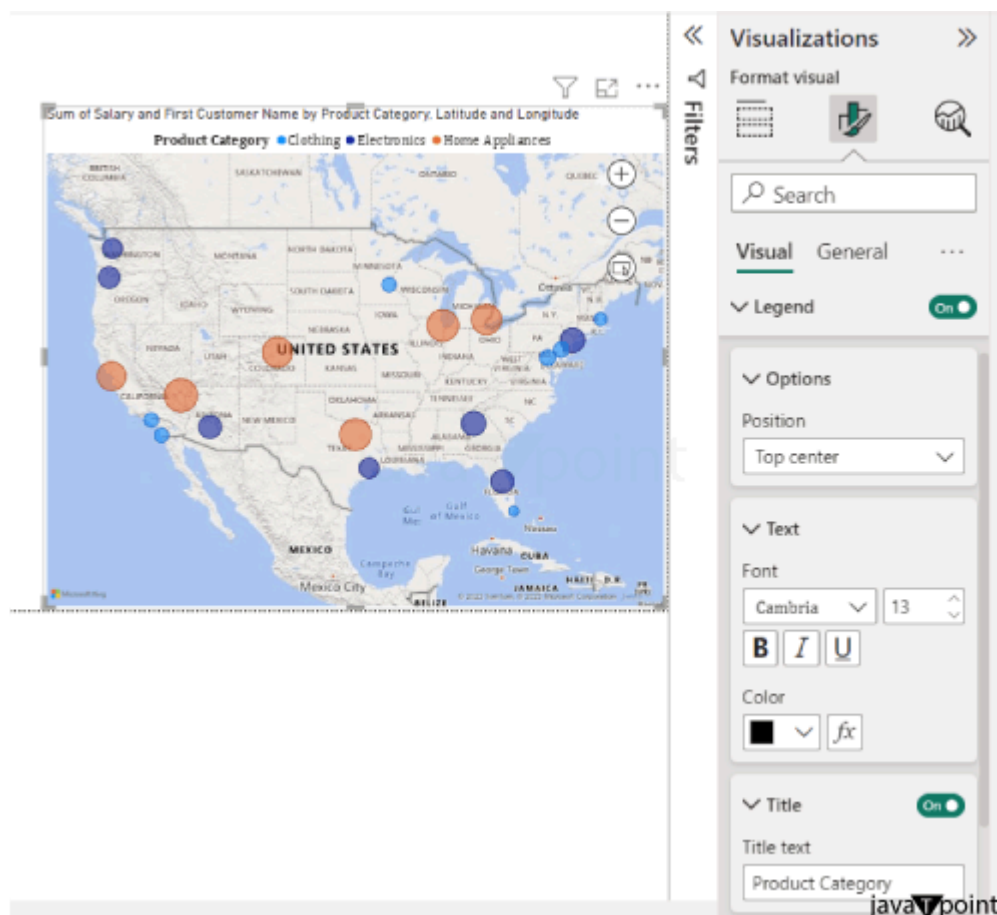
## Legends

Legends can assist you in categorizing your data and improving data analytics.

1. Choose "Legends" from the menu. The drop-down menu has various customizability choices, including "Options," "Text," and "Title."
2. The "Options" feature allows you to choose where the map legends are placed. There are alternatives such as top left, top center, and more.



3. It is possible to modify the color and font size of the legends in the "Text" element. For example, you can change the color of the legend of the Product Category to 13.



4. Click the "Title" property to modify the legends' titles. It has the same name as the column that you automatically added to the Legends section. Cambria can be used as the typeface name.

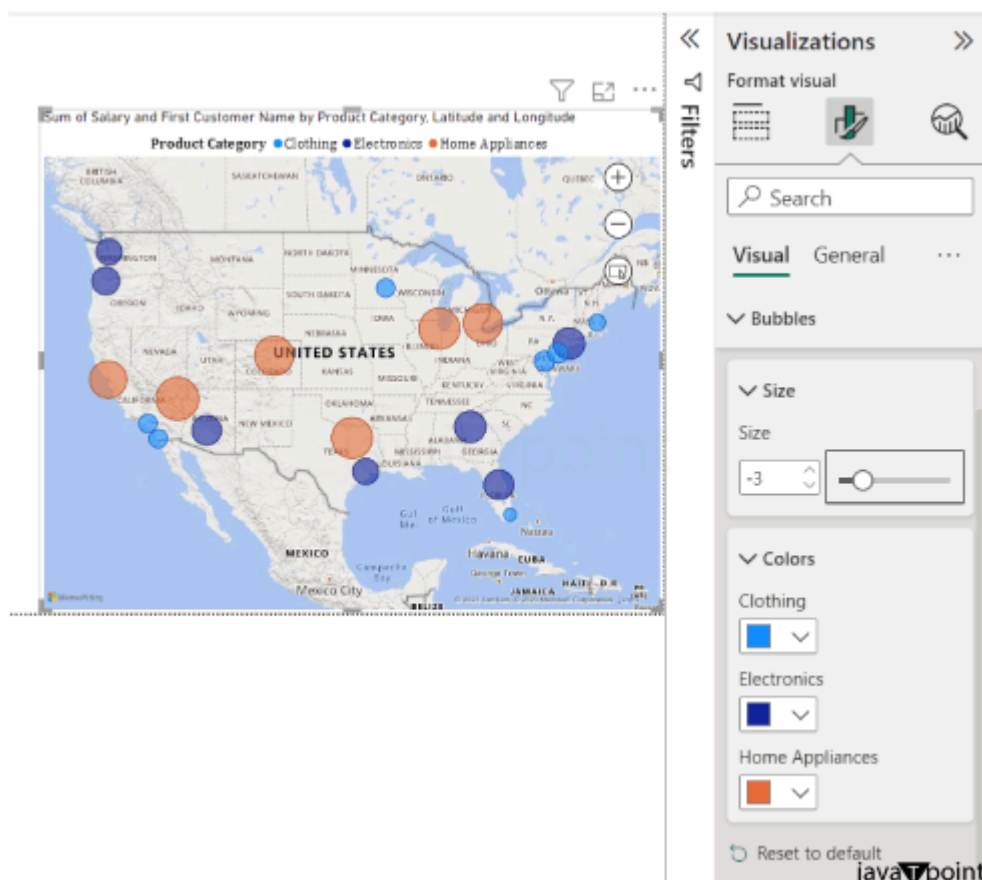
## Bubbles

In maps, bubbles are spherical symbols whose size corresponds to the numerical value of a specific column.

1. Choose "Bubbles" from the drop-down menu. Here, the two change options are "Colour" and "Size."

2. To suit your preferences, change the bubble size under the "Size" option. For example, you may change the size to -3px.

3. The "Colour" option presents a drop-down choice. Colors are categorized using the traits of the legend, and you can change the bubble colors to correspond to your classifications.



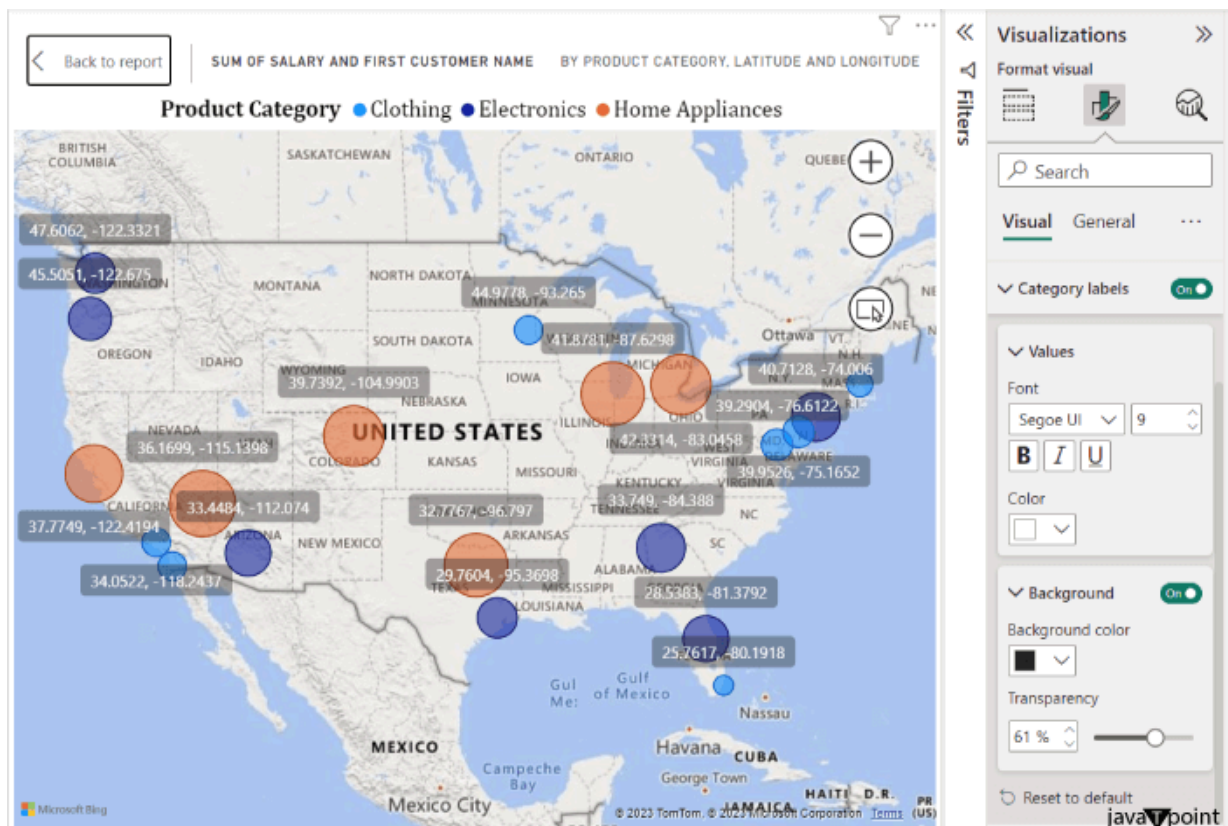
## Category Labels

Each data point in the category labels is accompanied by a tag that displays the values of the Location names.

1. Choose "Category Labels" from the menu for label customization options.

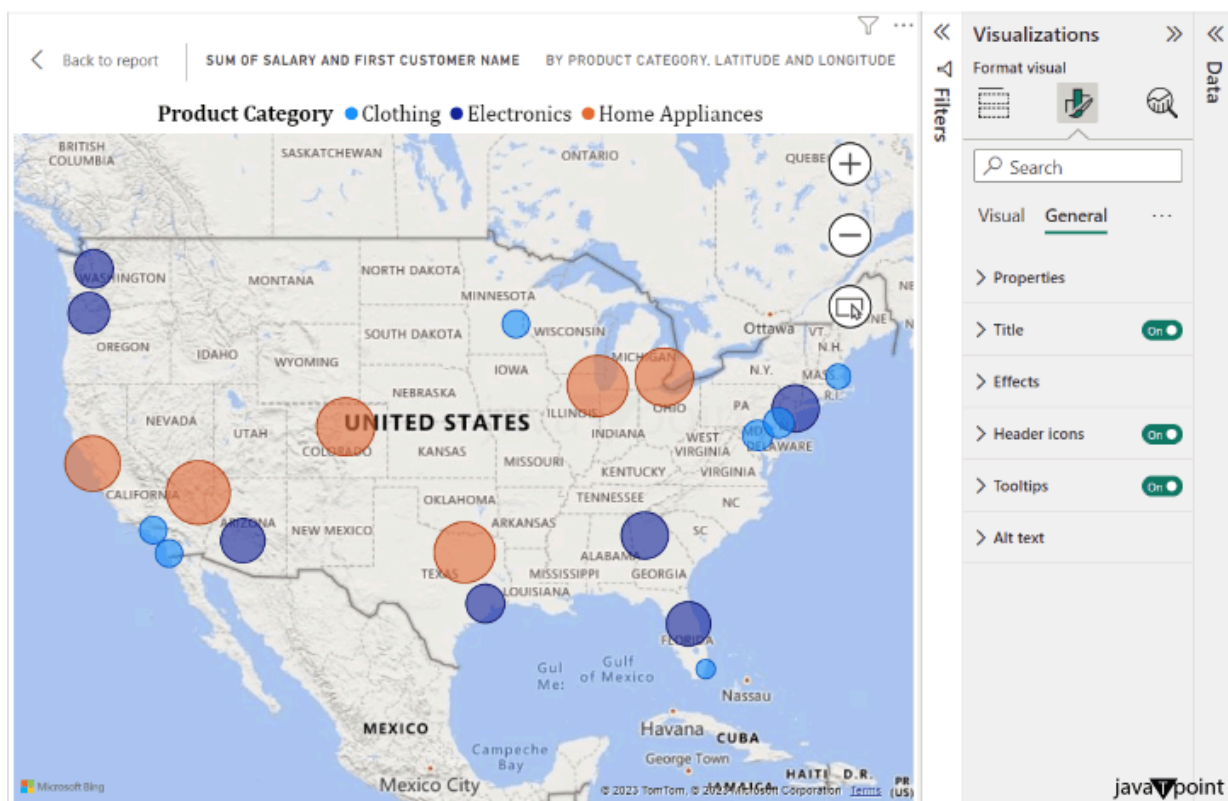
- Font: Change the size and script style of the category labels.
- Background: Colourize the background of the category labels.
- Transparency: Modify transparency, for instance, by setting it to 61%.





## General Formatting for Power BI Map Visuals

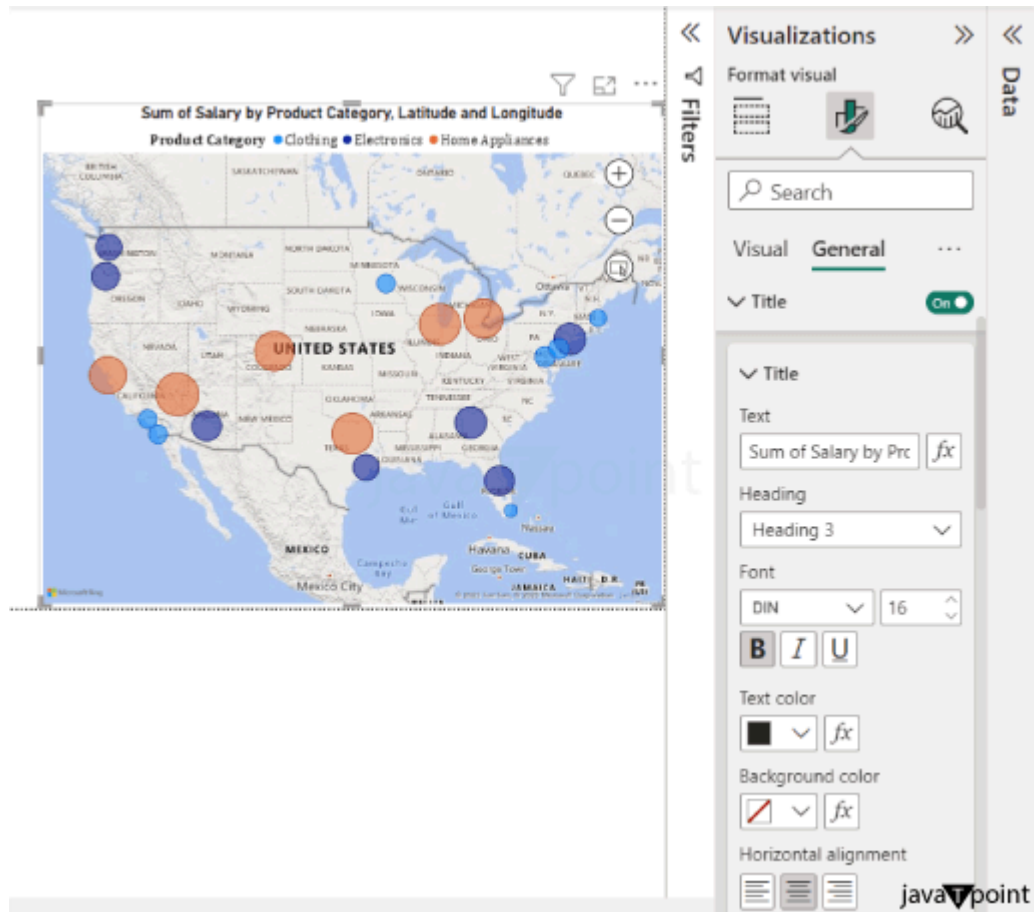
When making a map in Power BI, it's important to look at the various general formatting options available. With these settings, you may add titles, control tooltips, make header icons, and provide alternative text (alt text) for accessibility. Let's look more closely at each of these options.



## Title

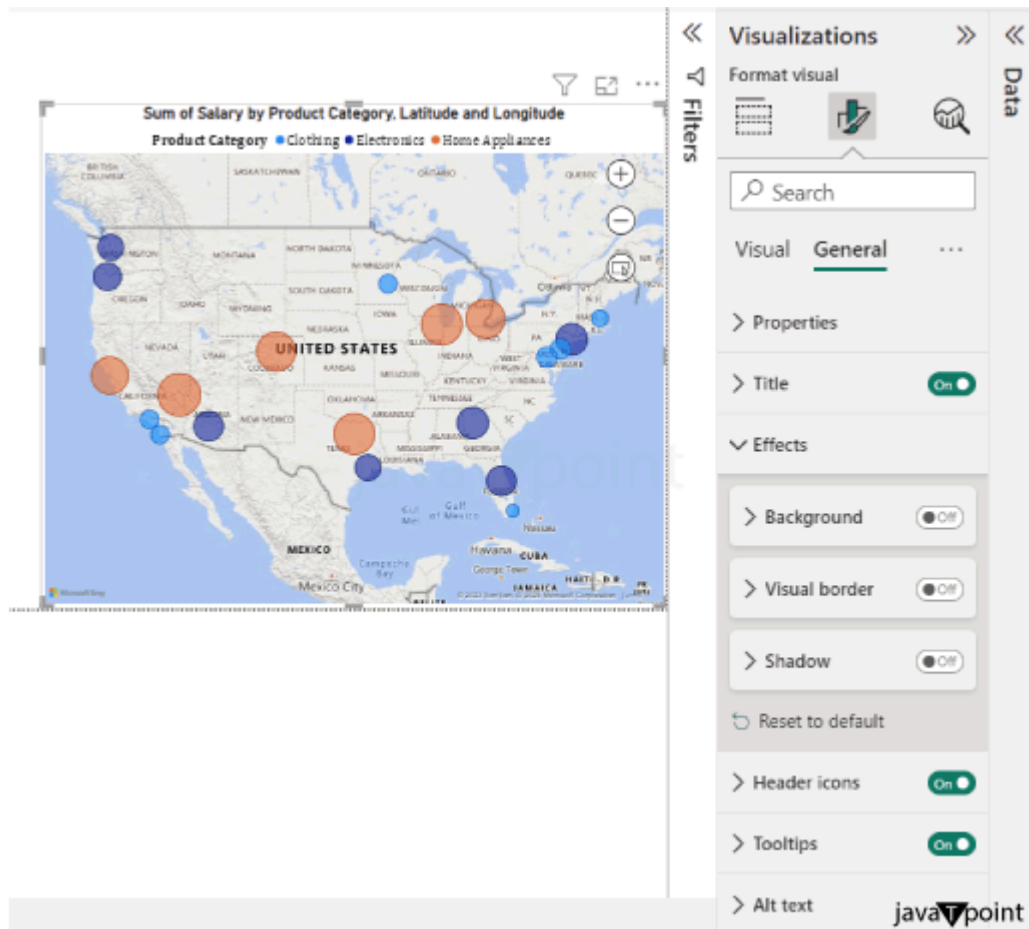
Adding a title to your map visualization improves its clarity and context.

1. From the menu, choose "Title". A drop-down menu will display options for you to customize your title.
2. Enter "Sum of Salary by Product Category, Latitude, and Longitude" to set the title in the "Text" field. In the picture below, you can see that the map now has a title. Additionally, the title can be changed by adjusting its size (for instance, to 16), font type, horizontal alignment (center), and boldness.
3. You may also change the color of the title. Under "Text color," select the color you wish to match your visualization's design.

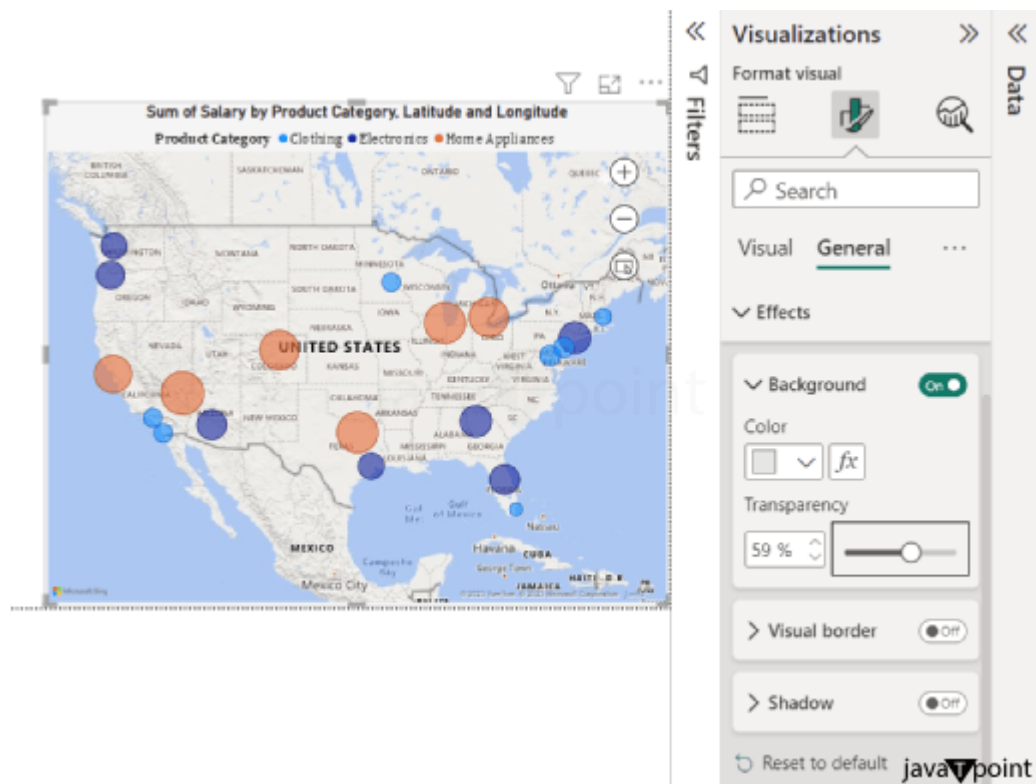


## Effects

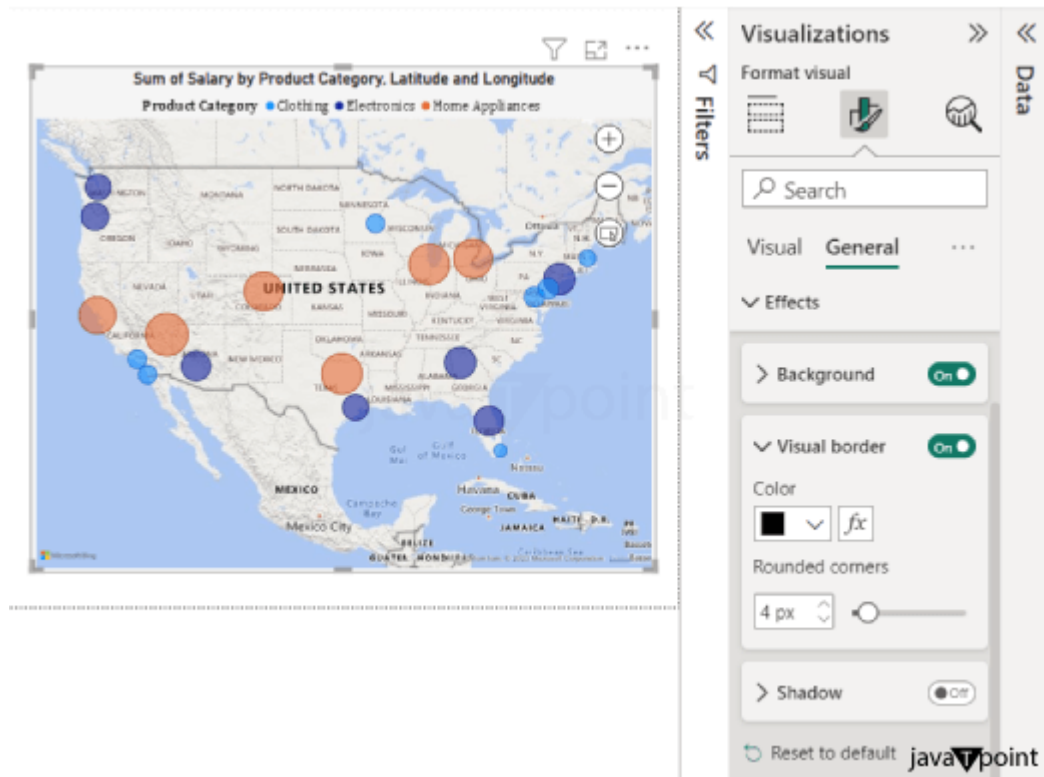
Three features are available in the "Effects" section: "Background," "Visual Border," and "Shadow," each with a distinct function.



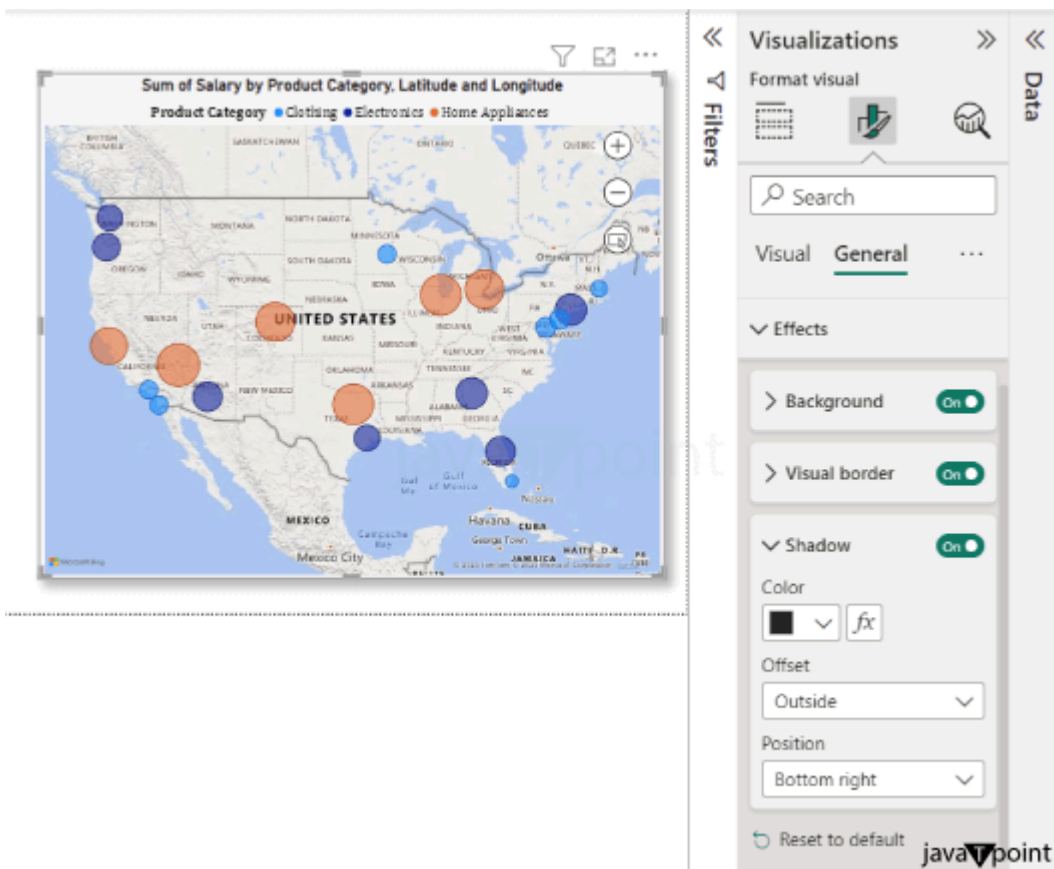
1. From the menu, choose "Background". The background color of the visualization is customizable. You might decide on black, for instance. The picture will mirror the map's dark background. Additionally, transparency can be altered to, say, 59%.



2. From the menu, choose "Visual Border." This will expand the visualization's 4px rounded corner radius and allow you to draw a border around it.



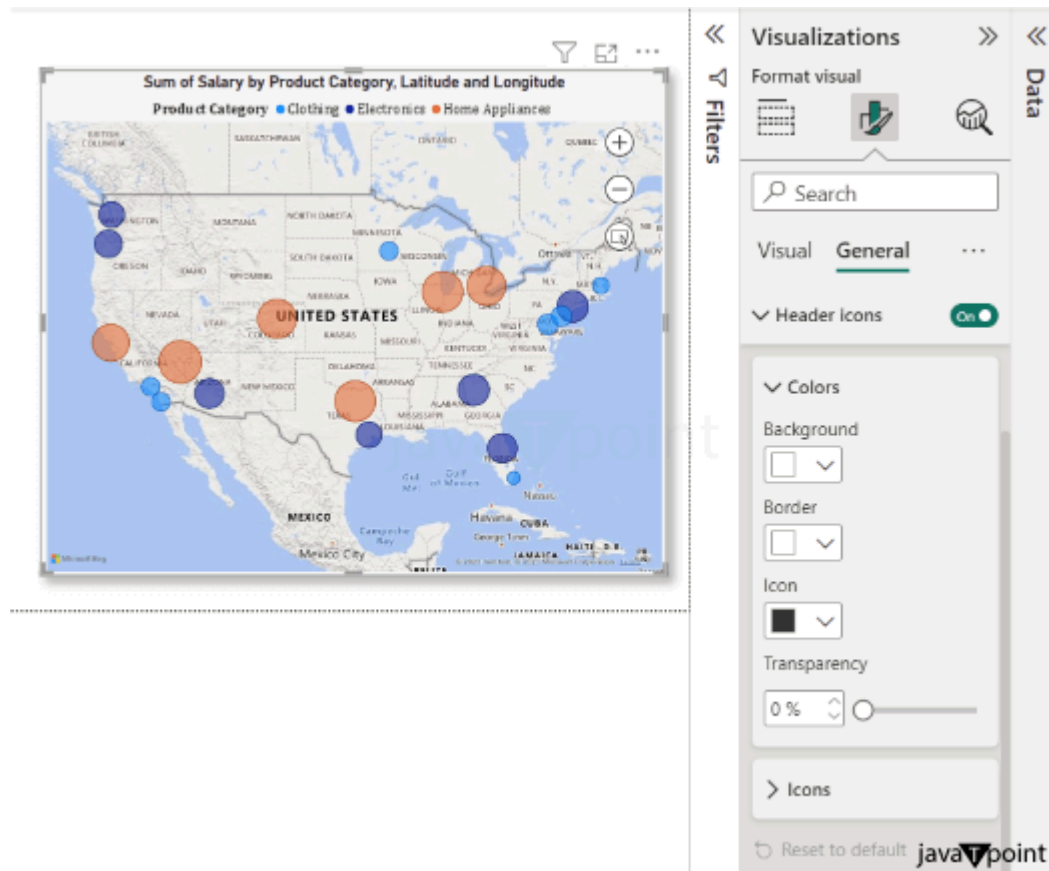
3. From the menu, choose "Shadow". This adds a shadow to the image, which you may then reposition to your preferred location.



## Header Icons

"Filter on visuals" and "More options" are the two options in the header icons at the top of the visualization. Click the "Header Icons" option to obtain a number of modification choices, including the ability to alter the background, border, and icon colors to fit your design better.

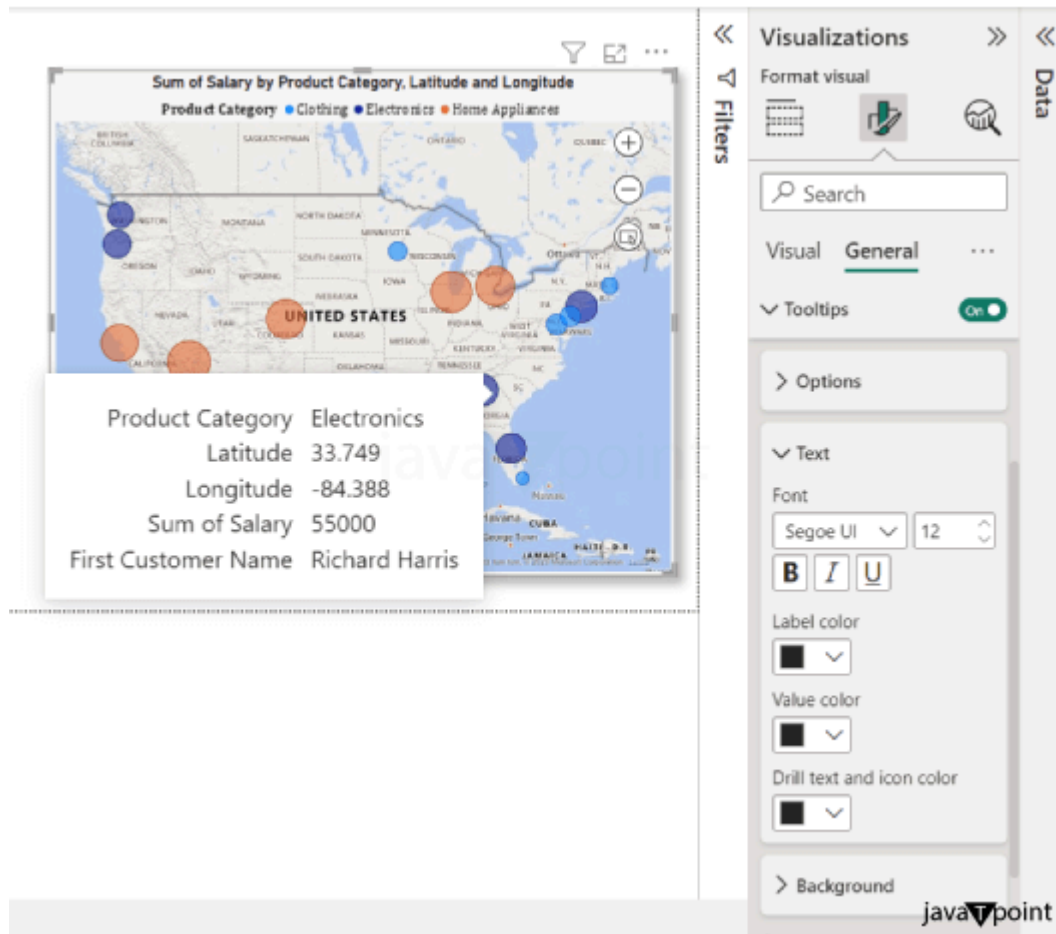




## Tooltips

You can see the fields by utilizing the helpful tooltip function and moving your mouse over the map.

1. Click on the "Tooltips" option, which offers three options: "Options," "Text," and "Background."
2. Under the "Options" property, you can configure settings to manage the look and functionality of tooltips. This allows you to regulate how users interact with the tooltips.
3. In the "Text" field, you can change the text and background colors to your preference. These options control the tooltips' appearance, making them aesthetically pleasant and easy to read.

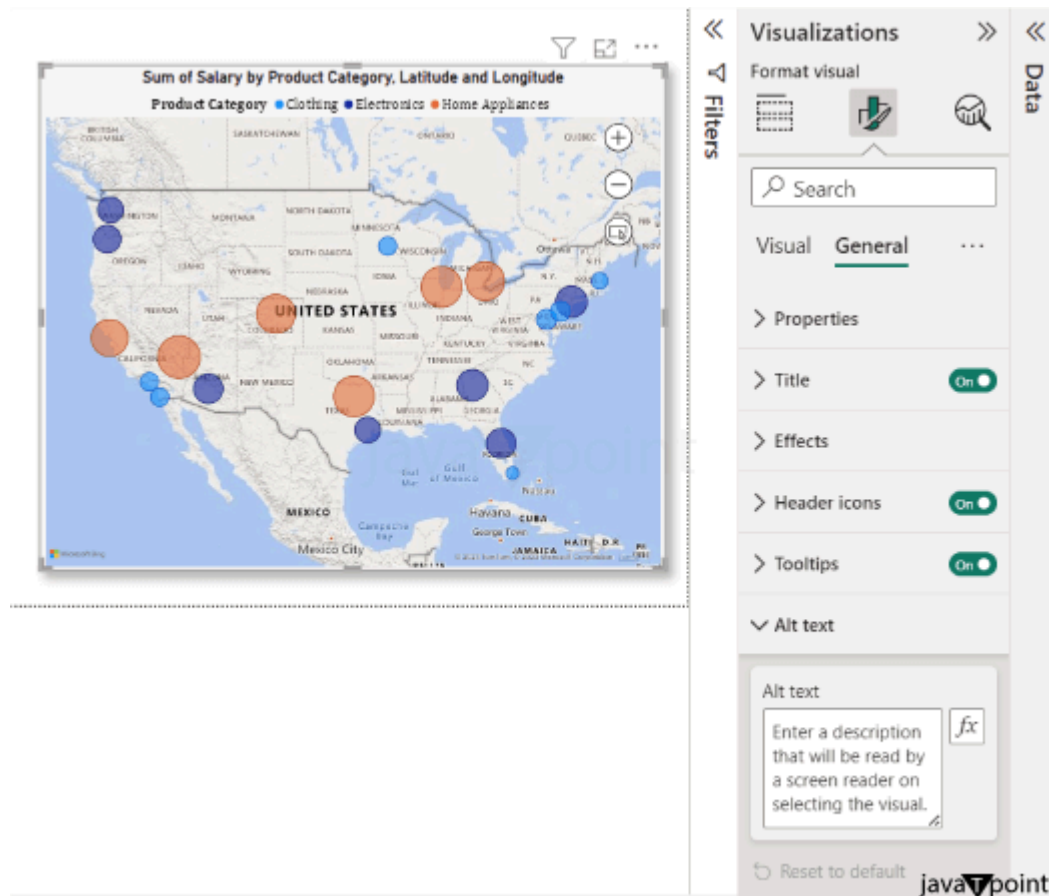


## Alt Text

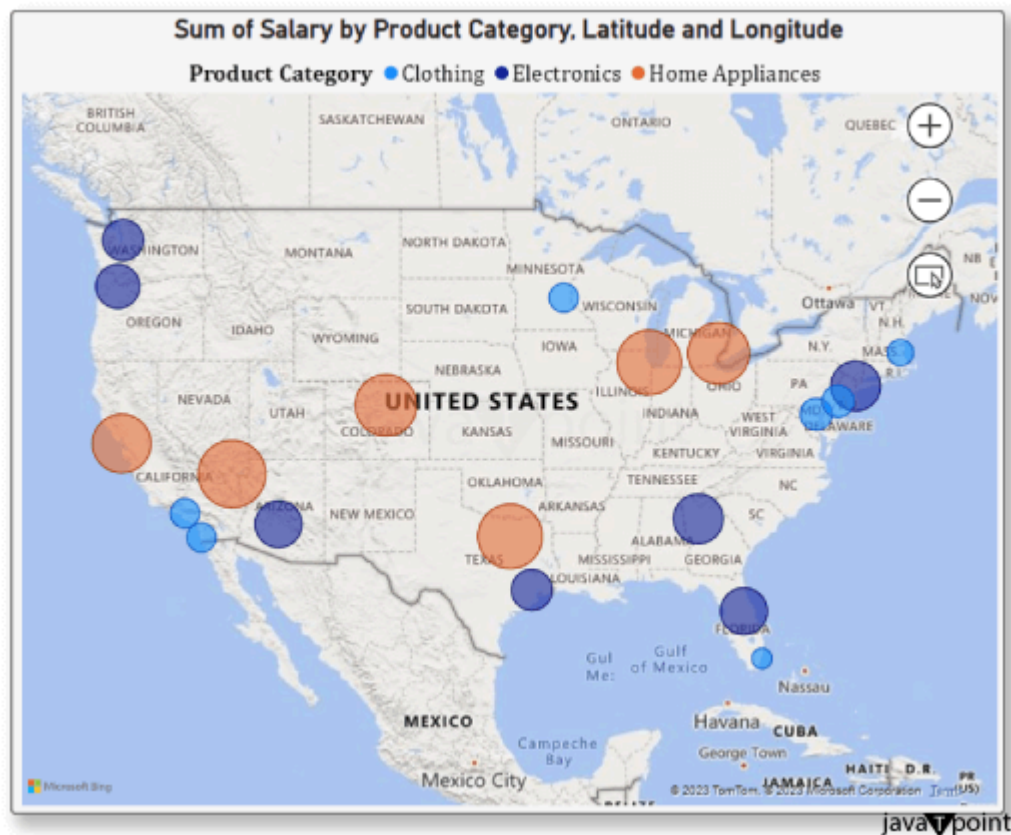
The "Alt Text" feature is crucial for accessibility as it allows blind or visually impaired users to comprehend the information using screen readers and other assistive devices. This option is only accessible if a screen reader is installed and active on the system (such as Narrator).

1. Select "Alt Text" and enter the text you want users of screen readers to hear when they see the visualization.

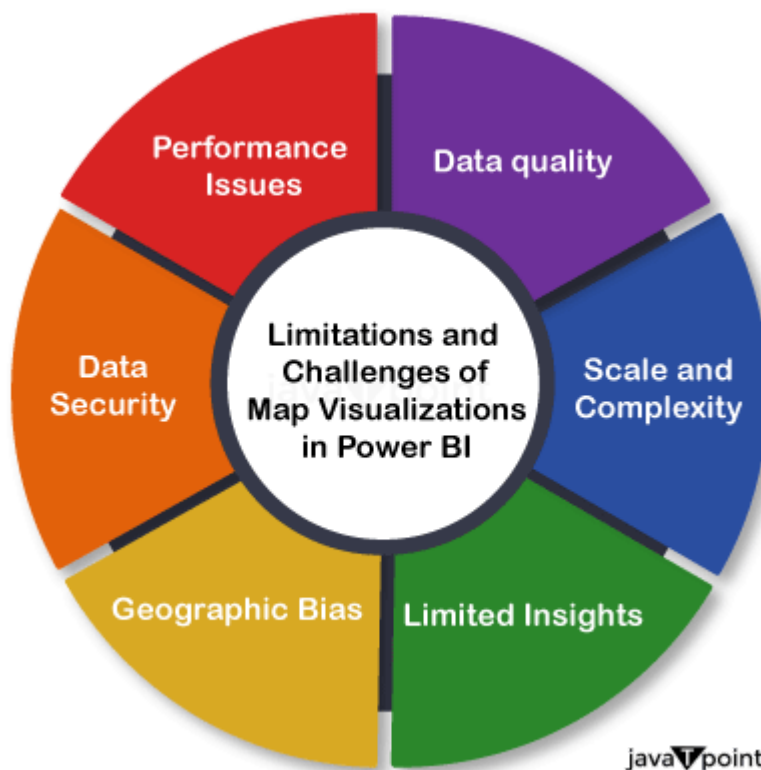




With the use of these visual and general formatting options, Power BI provides you with an extensive array of tools to customize the appearance and usefulness of your map visualizations. By carefully choosing these parameters, you may create inclusive and aesthetically pleasing visualizations, ensuring that a wider audience can access and understand your data presentations. The following image is the final visual which was created and customized using above steps



## Limitations and Challenges with Power BI Map Visualisations

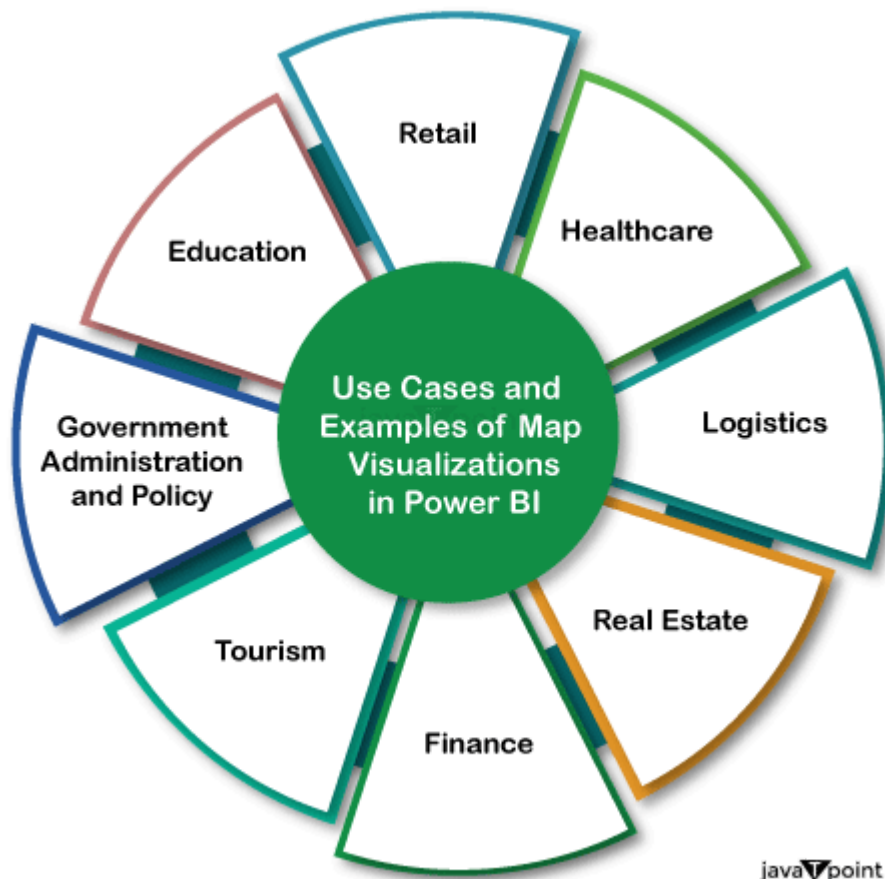


Map visualizations are powerful tools for data analytics. However, there are a few limitations and challenges to consider:

1. **Data quality:** For map visualizations, geographic data quality is essential. If the location data is accurate or updated, the information can be understood.

2. **Scale and Complexity:** Visualizing large datasets on a map with many data points can lead to clutter and confusion, making it challenging to draw meaningful conclusions.
3. **Limited Insights:** A map is the most straightforward tool for displaying geographical relationships. However, there may be better representations for other kinds of data, such as time series or complex relationships.
4. **Geographic Bias:** This bias makes locations with more data points seem more significant, which could lead to misinterpretations.
5. **Data Security:** Mapping sensitive data can result in privacy concerns when working with a person's location data.
6. **Performance Issues:** Power BI may occasionally experience performance issues when managing a large number of data points on a map.

## Use Cases and Examples of Map Visualizations in Power BI



Numerous industries can take advantage of Power BI's map visualizations. Here are numerous real-world examples and use cases:

1. **Retail:** A retailer uses map visualizations to assess the location of its stores and identify prospective growth and improvement areas. They can also keep track of regional sales progress.
2. **Healthcare:** Hospitals and healthcare professionals use map images to visualize patient demographics, illness outbreaks, and the allocation of healthcare resources. They can keep an eye on the spread of infectious diseases, such as COVID-19.
3. **Logistics:** Organisations that handle logistics use maps to optimize delivery routes, ensure on-time deliveries, and track trucks in real time. They can monitor the traffic situation and adjust their plans as necessary.

4. **Real Estate:** To show potential properties and tell clients about the neighborhood and services, real estate firms employ map visualizations. They can also look at trends in local home markets.
5. **Finance:** Banks and other financial institutions utilize maps to display the locations of their branches, ATM distribution, and client demographics. This aids strategic planning and market segmentation.
6. **Tourism:** trip companies use maps to make interactive trip guides that show accommodations, dining options, and areas of interest. This helps travelers plan their trips with greater knowledge.
7. **Government Administration and Policy:** Government agencies utilize maps to show data from censuses, elections, and infrastructure development. When making judgments based on information that impacts citizens, it's a helpful tool.
8. **Education:** Educational institutions can utilize education maps to present data on student demographics, campus locations, and enrollment numbers. This may be beneficial for recruiting students and allocating resources.

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